

Department of Physical Science
Faculty of Applied Science
University of Vavuniya

Parallel Computing - IT3243

Tutorial No 01: Introduction to Parallel Computing

1. Define the term “**parallelism**” in your own words.
2. Explain briefly how **concurrency** differs from **parallelism**.
3. What is the difference between Sequential and Parallel Computing?
4. Describe each of the following in the parallel computing: **data level parallelism**, **task level parallelism**, and **instruction level parallelism**.
5. Explain why **parallelism** is important to support the computational needs in the context of modern hardware.
6. Explain the differences between **high performance computing** and **big data computing**.
7. What is a **processor/core**?
8. List types of **processors** used in computers.
9. What do you mean by the term **thread** in the context of parallel computing explain how **thread-level parallelism** is incorporated in parallel computing.
10. Write features that are dependent of **performance of the CPUs**.
11. Explain the **two** main types of **parallel systems**.
12. Distinguish the terms **concurrent**, **parallel** and **distributed systems**.